



MINING DEALER BEST PRACTICE

Performance Evaluation (Strategic)

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

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1.0 Introduction

The principal mission of the maintenance organization is the preservation of the primary function of the equipment throughout his productive life. All routines and processes required to maintain the equipment function must be established in order to minimize equipment downtime and the resulting production losses.

Caterpillar Global Mining Maintenance & Repair process contains ten distinctive or specific sub processes that are organized in five principal steps:

Detect the Maintenance & Repair Needs

Plan and schedule the Maintenance Events

Execute repairs efficiently and effectively

Evaluate results

Correct or Improve through the application of direct known actions and/or by Continuous Improvement.

As it can be seen one of the key steps is the evaluation of results. The work is not completed at the end of the execution of the repairs. Effective management requires measuring performance or result, comparing these results with targets and goals, discovering or identifying the problems or sub-standard condition, focusing prioritized actions to improve, correct or ensure sustainability of gains already obtained.

This Strategic Best Practice is dedicated to describe the overall process and routines of the M&R Performance Evaluation Process and how it will together with the rest of the M&R processes support your Maintenance organization to achieve the expected results.

2.0 Best Practice Description

The demanding goals of today's Mining Industry has motivated and reinforced the continuous search of operating efficiency in all areas of the production process. Mobile equipment plays an important role in the achievement of these goals, and their ability to perform according to the specifications relies on the observation of three major factors: Maintenance, Design and Application. Each of these three areas has to be considered and managed effectively to optimize the productivity of the equipment.

A well designed and effectively implemented Maintenance System considers and manages all the variables related to these 3 major factors, Maintenance, Design and Application through a series of various specific processes with specific outcome that supports the final results of the mining equipment fleet.

"You can't Manage what you don't Measure", is an old and well known management expression. It reinforces the unavoidable need for measuring, to detect if the outcome of any actions or processes is getting better or worse.

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Performance evaluation is the process dedicated to measure, identify the areas of opportunities and/or problems, provide timely information to the process owners to support and facilitate the identification of corrective actions to enable effective management of the M&R operation.

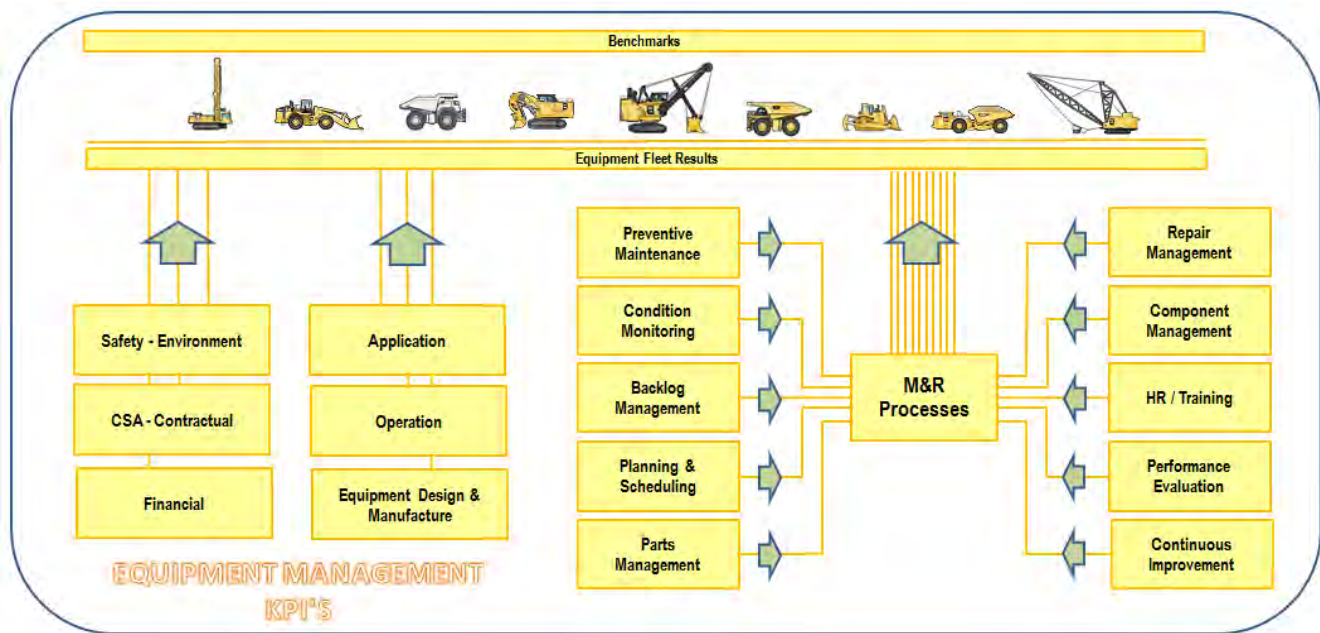


Illustration 1: The end result or overall performance of the Maintenance & Repair operation is a combination of the contributions or outcome of the various processes that compose the M&R process model.

The explanation of the Evaluation Process will be addressed in this Best Practice document dividing the overall process in six general steps, Strategy Definition, Data Collection, Transform Data into Information, Interpretation – Analysis – Action Plan, Implementation – Validation – Learning and the last Process Management.

Performance Evaluation High Level Process



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Step 1: Strategy Definition

In this section the definition of 6 basic questions are address: WHAT – FOR WHAT – HOW – WITH WHAT – WHEN – WHERE ARE WE.

WHAT: what is our mission statement ... what do we supposed to do.

FOR WHAT: What are our goals, objectives, targets of the overall operation and sub processes.

HOW: Description of procedures and methodology on how we are going to do it.

WITH WHAT: What are our resources to execute the tasks and achieve the mission goals and objectives.

WHEN: when are the tasks performed, with what frequency, etc.

WHERE ARE WE: A necessary step to be able to manage the process. In this step we define the reporting system.

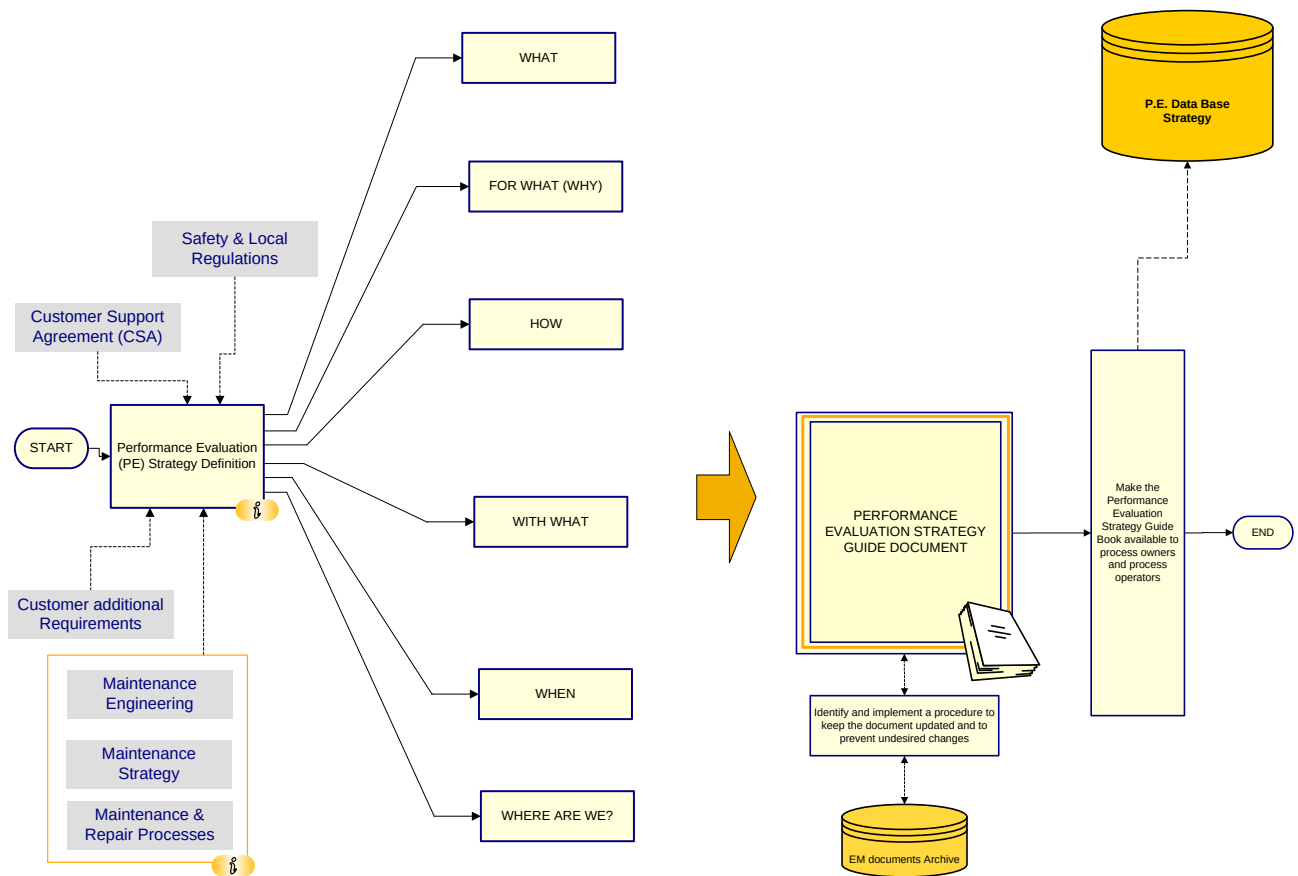


Illustration 2: Building the Strategy for the PE Process

The main output of this stage is the documented strategy for the Performance Evaluation Process. This is one of the most vital steps in any process, the definition of the strategy, definition of our Mission, Goals and Objectives, How we are Going to Do It (working procedures), Resources, best timing to perform the various routines and the way we will report where we are with respect to our goals and targets.

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The strategy is the map, the “blue print” to follow to achieve the desired outcome. For this reason the strategy must be documented, archived and made available to all personnel in the organization. The whole organization at a different level of involvement has to have easy access to the strategy to understand the direction, challenges and goals we are facing.

Step 2: Data Collection

In this step all data sources access are activated and the required inputs defined in the strategy, in the “KPI’s Matrix”, are collected. The data reconciliation is also considered in this phase.

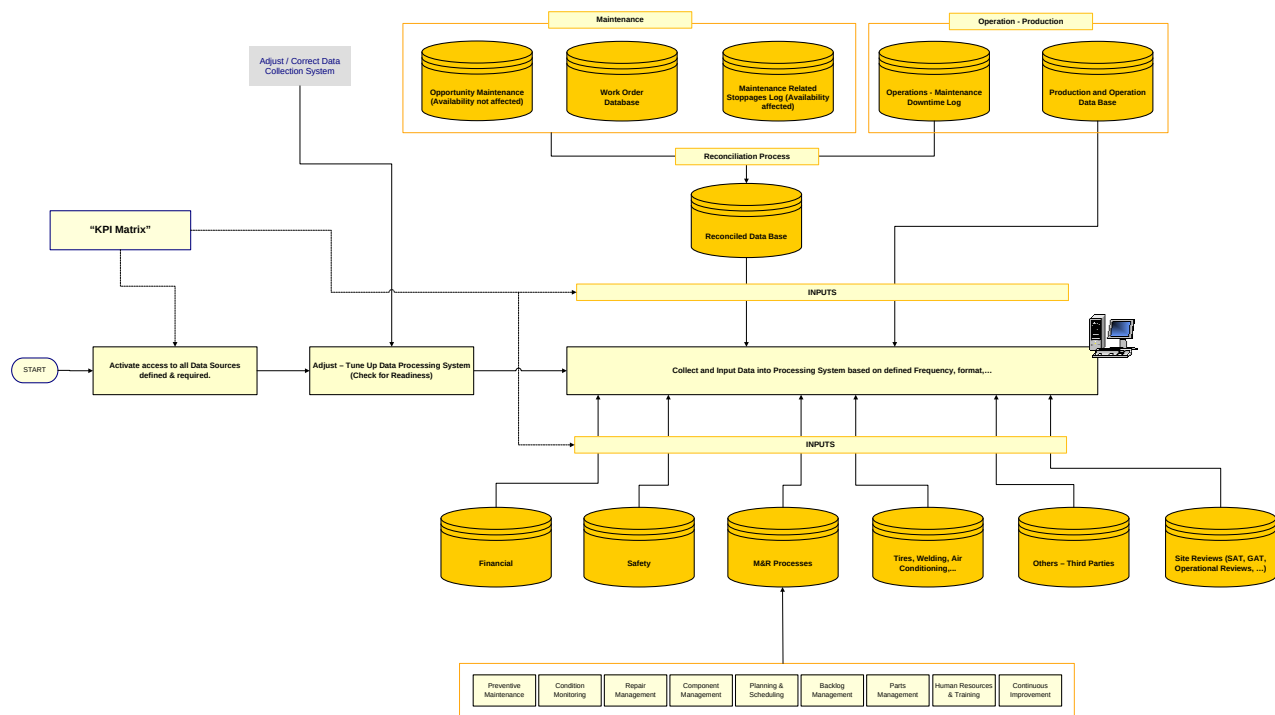


Illustration 3: Opening the “access” to all data source / Data Bases.

Step 3: Transforming Data into Information

After the data gets collected and reconciled the various KPI’s are calculated following the formulas and calculation methodology defined in the strategy and the first stage of structuring the management reports (“Phase I” is addressed). Here we start transforming the data into meaningful information.

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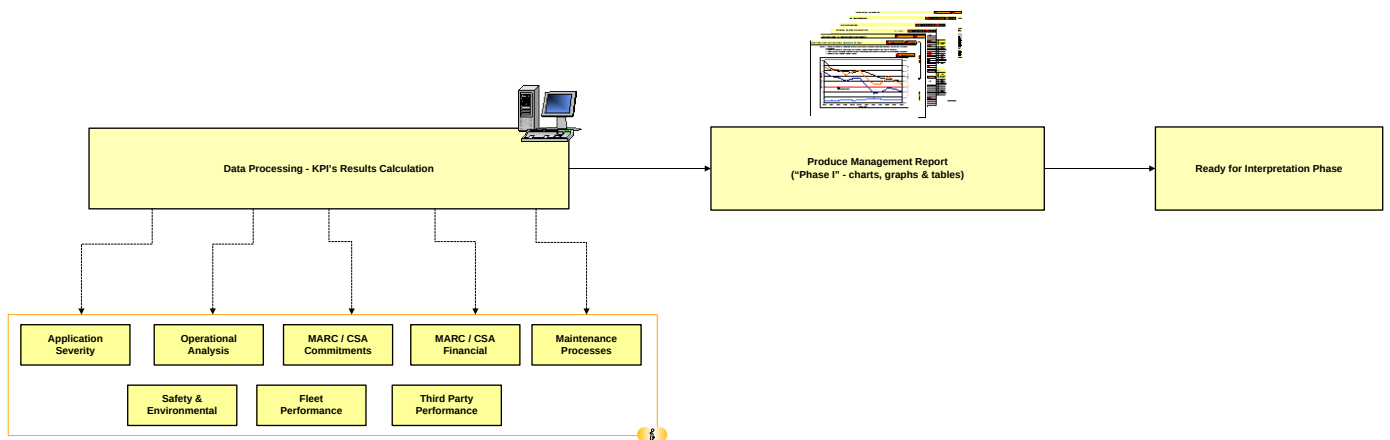
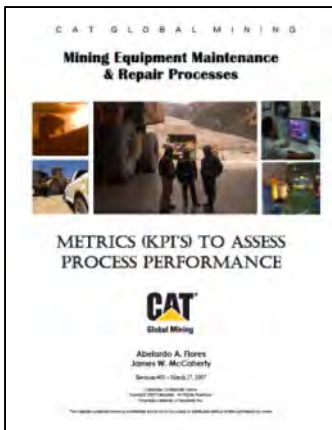


Illustration 4: KPI's Calculations and first step of building the management reports

Note 1: We call “Phase I” management reports to the reports that are in the process to be completed that only includes the results of the metrics calculation presented in Graphs, Tables or other forms. At this stage the interpretation and corrective actions are not considered.



Note 2: For KPI's calculation formulas, procedures and recommended targets & benchmarks we recommend consulting the CGM booklet “Metrics (KPI's) To Assess Process Performance.”

This document contains the metrics recommended to assess the overall Maintenance & Repair operation (Top Tier Metrics) and the individual processes involved (Process Metrics).

Step 4: Interpretation, Analysis & Action

This step is dedicated to the interpretation of the KPI's / reports, analysis and detection of the opportunities to improve determining the corrective actions needed. The interpretation of the metrics / management reports is a task that must be taken by the responsible department for structuring the reports and the on-site process owners and managers. It is a collective responsibility.

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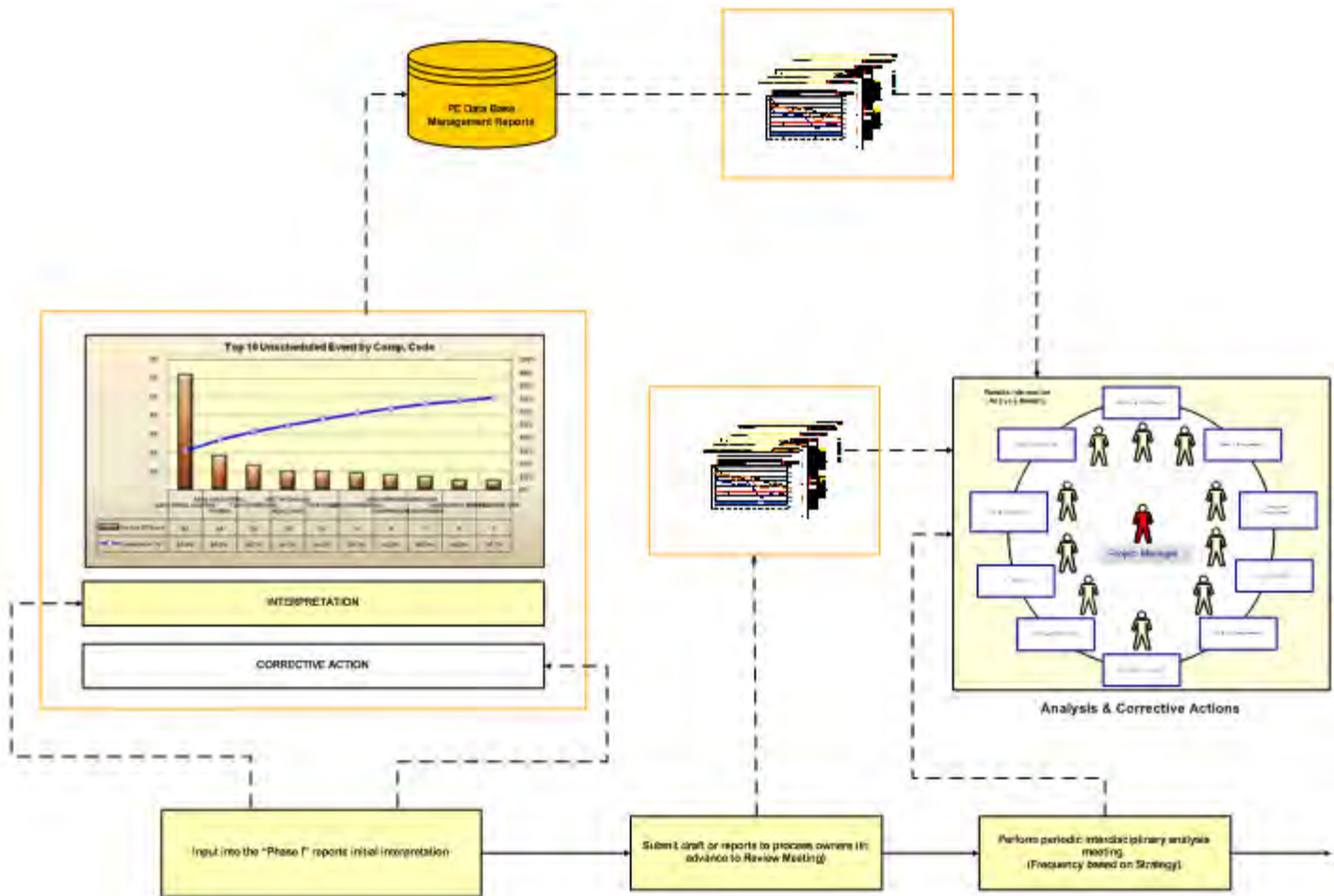


Illustration 5: The reports (Phase I) is sent to the on-site organization process owners in preparation of the final Analysis – Corrective Actions identification meeting.

The management reports are normally being prepared by the Planning or Maintenance Engineering department and then send to the process owners for their review and contribution to the final analysis and interpretation. The organization gets together in a formal meeting to review, discuss and find the opportunities to correct or improve.

The participation and feedback of the on-site organization is important first to consider “first hand” information / feedback to support the interpretation of the KPI’s and secondly to get the appropriate commitment from everybody to move forward with the corrective actions to implement.

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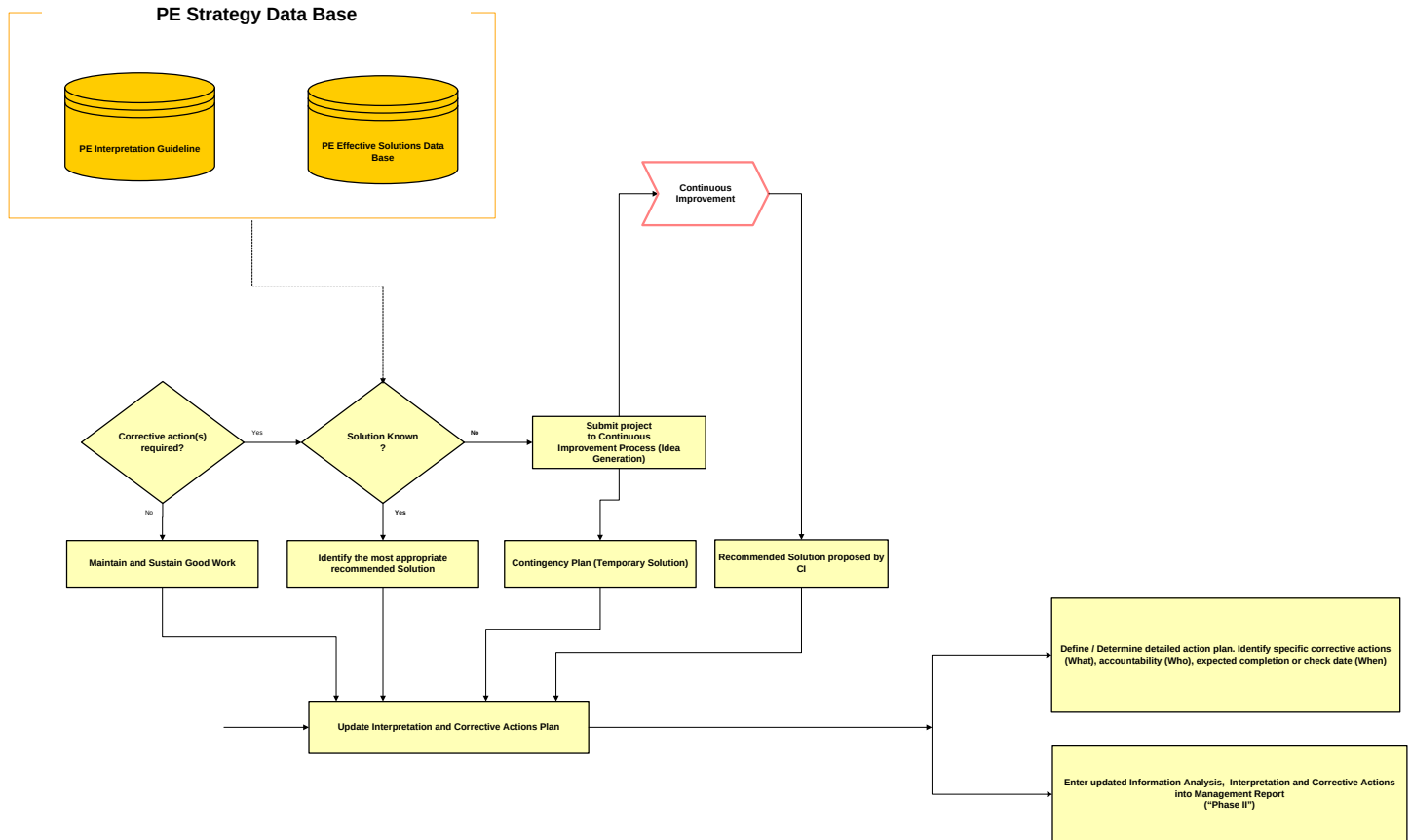


Illustration 6: The organization identifies the opportunities and corrective actions to apply.

As a result of this step the final interpretation with the correspondent corrective actions are included in the management reports and in the 3W action plans. The Management Reports and the 3W action plans “must speak to each other”. We measure – Interpret – Identify Opportunities – and Act, this is a chain of events that must be very clear to each members of the service crew.

Step 5: Implementation, Validation and Learning

This phase is dedicated to the implementation of the corrective actions; follow up of the results, introduce adjustments if necessary and finally learning from the results. We have to ensure we always learn from the good and specially the bad results of our management actions.

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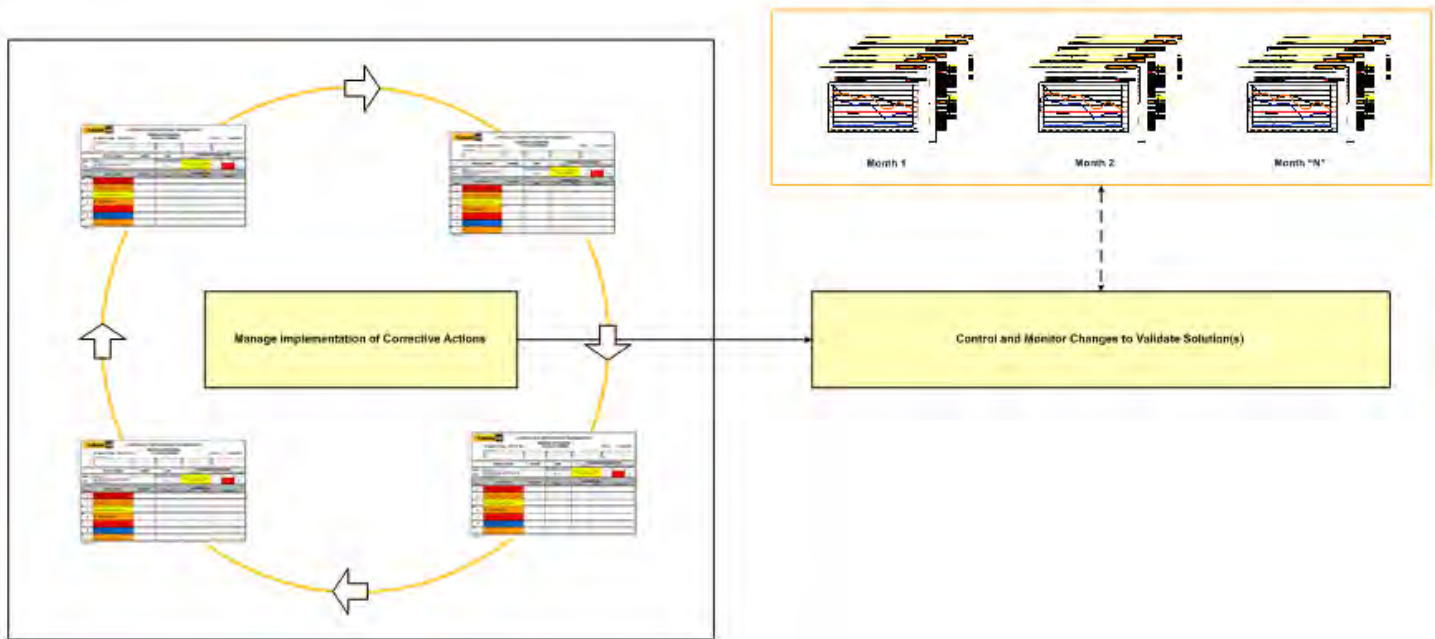


Illustration 7: The corrective actions are being implemented and the results monitored.

The corrective actions effects are closely monitored through the KPI's results and direct observations of the process owners. Both positive and not that positive effect are being captured to enrich the Interpretation Guidelines and Effective Solutions Data Base (Continuous Learning).

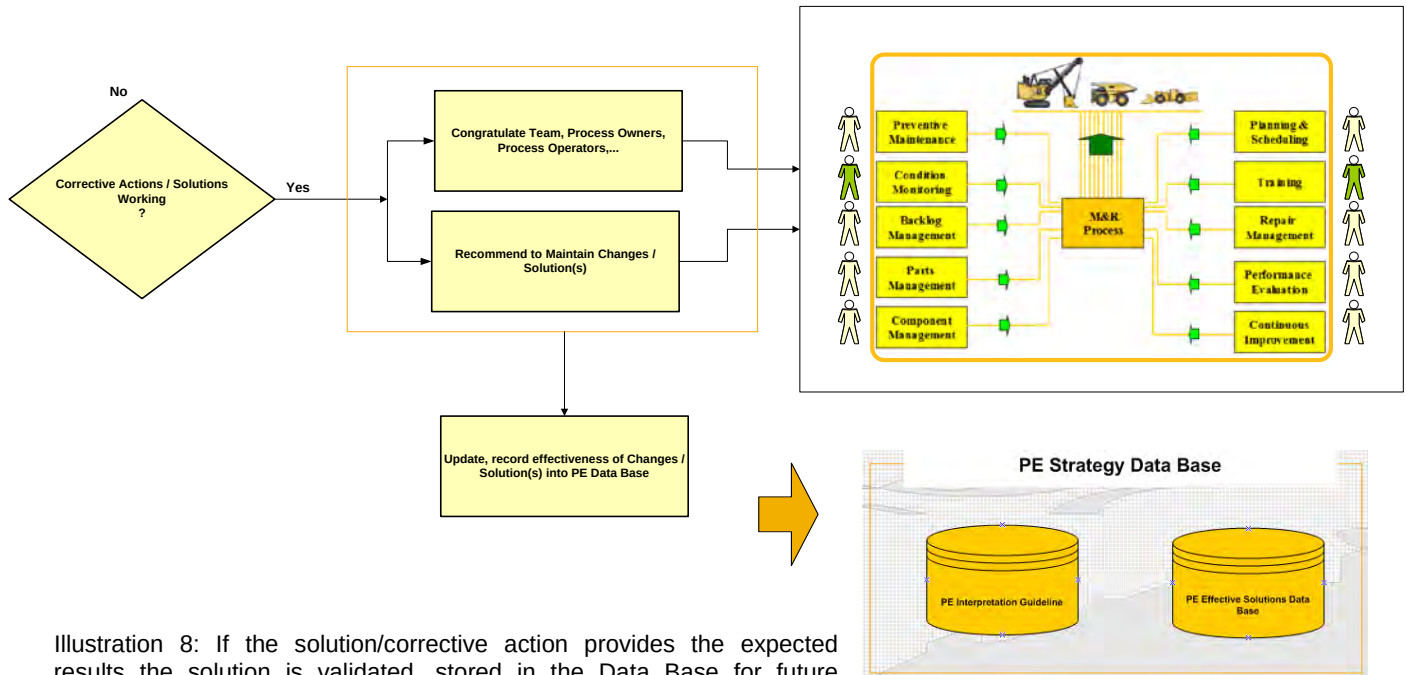


Illustration 8: If the solution/corrective action provides the expected results the solution is validated, stored in the Data Base for future references, the team is congratulated and the new procedures/solution is incorporated as a regular operating procedure to maintain the gains.

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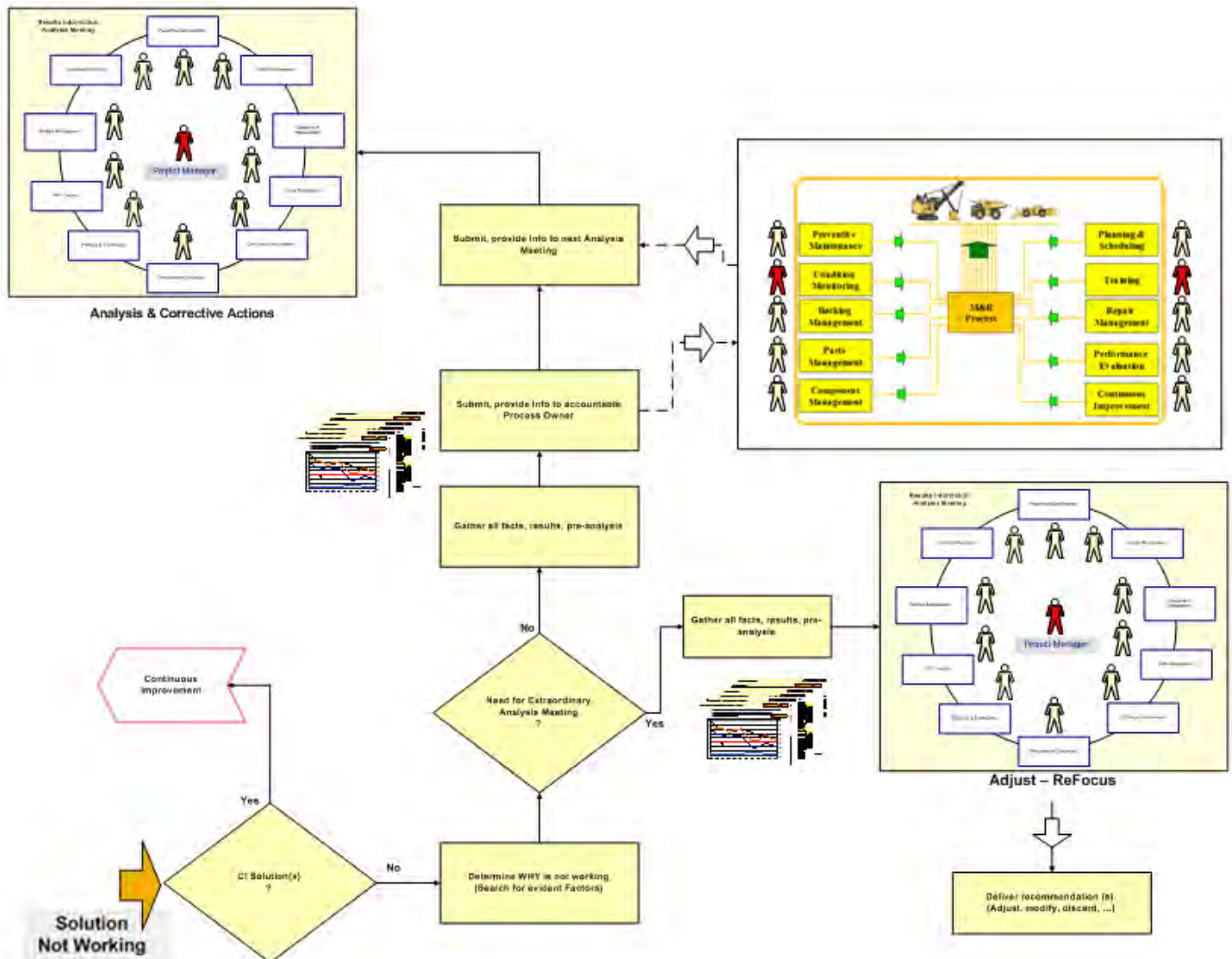
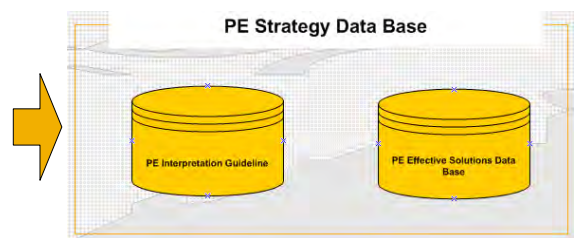


Illustration 8: If the solution/corrective action doesn't work, the team has to define the most appropriate course of action: involve the team of CI if it is the case, make an extraordinary analysis meeting to adjust or refocus the corrective actions and or gather prepare all information for the next Equipment Management results analysis meeting.

In the same way for those solutions that doesn't work we need to gather all related information and store it in the Data Base for future reference. We need to learn from our mistakes.



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Step 6: Process Management

This stage is dedicated to manage the necessary actions to control the process and ensure the "expected outcome(s)" are met. It's like managing a factory, keeping the processes working to meet the production targets. The main steps in managing a process are:

Define the Outcome (*Product*) → Goals & Objectives → How we are going to Measure (*KPI's*) → Acceptability Criteria (*Targets*) → Measure → Interpret → Identify corrective Actions → Implement → Follow Up



Illustration 9: The Final Results (Equipment Fleet Performance) is the result of the collective contributions of the involved processes; in this case the 10 M&R core processes

The 9 steps above applies to the overall M&R process as well as to the 10 individual key functions, ...the final result overall outcome is the result of all the contributions of each of the 10 M&R elements. So, if we want to keep the factory running and producing at the design level we need to manage each of these 10 fundamental elements.

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M&R Processes Goals-Outcomes and KPI's Definition

We measure because we want to be able to apply the correct actions, manage the processes and ensure the achievement of a specific outcome. The identification of the outcome(s) of a particular process is the first step in defining the most appropriate metrics to use.

The Performance Evaluation process takes the lead in linking the outcome of overall M&R operation and the outcome of each of the M&R processes with the recommended metrics.

M&R Outcomes and KPI's – Top Level Outcome

First to define is the overall outcome the M&R organization has to aim for.

Top Tier Results

The final performance of the equipment fleet is our goal and ultimate outcome. We can summarize this outcome or goal as:

“Maintain the equipment in optimum conditions to support the current and future production goals and the operation cost”

The following measures are “Top Tier” performance metrics or Key Performance Indicators (KPI's) that enable management to assess at a high level the results of the mobile equipment fleet.

- **Production Plan Compliance**

This metrics is a comparison between production plan and actual production tonnage, expressed in percentage. Even though this metric relates to the customer meeting the production, it is also important for the maintenance organization to understand the core business and their participation in the final production value chain

$$\text{Production Plan Compliance} = \frac{\text{Actual Production Tonnage}}{\text{Production Plan Tonnage}} \times 100 (\%)$$

- **Utilization (Asset and Availability)**

Utilization or operating hours is the metrics that will provide an indication of revenue (for internal business) and level of utilization of the equipment to support the production plan.

Asset Utilization is the proportion of time that a machine is operating (operating hours) divided by the total calendar time in the period, expressed as a percentage.

$$\text{Asset Utilization} = \frac{\text{Operating Hours}}{\text{Total Calendar Hours}} \times 100 (\%)$$

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Another metric to measure the utilization is **Utilization of Availability**. This metric is defined as the ratio of operating hours to the time the machine is capable of functioning (available hours).

$$\text{Utilization of Availability} = \frac{\text{Operating Hours}}{\text{Operating Hours} + \text{Standby Hours}} \times 100 (\%)$$

▪ Availability (Physical, Contractual and Index)

The availability metrics provide the total equipment hours that the Maintenance and Repair organization is giving to the production department. This is a good indication of the maintenance organization capability to support the production plan. There are three types of availability which are normally used.

Contractual Availability

Contractual Availability is the ratio of time that a machine is capable of functioning in the intended operation (available hours) to total hours in the period under consideration (typically either based on calendar or scheduled hours), expressed as a percentage. The calculation of available hours is not a pure calculation since the result is amended by downtime hours that are specifically excluded or limited by the terms of the contract.

$$\text{Contractual Availability} = \frac{\text{Total Hours} * - \text{MARC Downtime Hours}}{\text{Total Hours} *} \times 100 (\%)$$

*Note: "Total Hours" are typically either based on calendar or scheduled hours depending upon the calculation methodology defined in the agreement or customer expectations.

Availability Index

Availability Index is defined as The ratio of MTBS (average shutdown frequency) to the sum of MTBS and MTTR (average shutdown duration), expressed as a percentage.

$$\text{Availability Index} = \frac{\text{MTBS}}{\text{MTBS} + \text{MTTR}} \times 100 (\%)$$

The production department normally looks at Physical Availability that becomes also one of the top goals for the maintenance department. However, the maintenance department in order to have a more effective availability management tool use the Availability Index, that provides a performance result that is not influenced by the utilization of the equipment and also it provides

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a direct link to the frequency of the stops (MTBS) and the duration of the M&R events (MTTR) or turnaround time.

▪ **Mean Time Between Shutdowns (MTBS)**

The metric is defined as the average operating time between machine stoppages. It is a function of the average frequency of equipment downtime events, expressed in hours.

$$\text{MTBS} = \frac{\text{Operating Hours}}{\text{Number of Shutdowns}} \text{ (hours)}$$

▪ **Mean Time To Repair (MTTR)**

The metric calculates average downtime hours (including delays and response time) required to complete repairs, expressed in hours.

$$\text{MTTR} = \frac{\text{Total Downtime Hours}}{\text{Number of Shutdowns}} \text{ (hours)}$$

▪ **Scheduled Downtime**

The metric measures the percentage of total downtime hours performed in a given period that have been planned and scheduled.

$$\% \text{ Scheduled Downtime} = \frac{\text{Scheduled Downtime Hours}}{\text{Total Downtime Hours}} \times 100 \text{ (\%)}$$

▪ **Maintenance Ratio**

Maintenance Ratio is a dimensionless ratio of maintenance and repair man-hours to machine operating hours. It is an indication of the amount of effort required to keep equipment in service, as well as the efficiency with which labor is deployed and the effectiveness of the workforce in carrying out its duties. Maintenance Ratio can be calculated as either “charged” or “overall.” “Charged” Maintenance Ratio considers only work order man-hours (direct labor). Repair Shop, Component Rebuild Center, and labor is not included in the calculation. “Overall” Maintenance Ratio includes all the elements of “charged” Maintenance Ratio plus staff, supervision, and idle time.

Maintenance and repair man-hours are obtained from the work order. It should include actual time spent working on all forms of maintenance, repairs, and modifications, as well as inefficiencies that result from inspection and diagnostic time, or any delay for bay space, parts, tooling, literature, repair support equipment, decision making, etc.

Operating hours are obtained from machine service meter readings and should include production delay hours.

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$$\text{Maintenance Ratio} = \frac{\text{Maintenance and Repair Man – Hours}}{\text{Operating Hours}}$$

▪ Top Problems Summary

Top Problem Summary is the distribution of problems affecting a fleet of equipment ranked in terms of MTBS, MTTR, Impact on Availability and Costs. All mining support operations have limited resources. The most successful operations are those that have a clear understanding of the problems and issues facing them to focus their efforts and allocate the appropriate resources.

$$\text{MTBS} = \frac{\text{Operating Hours}}{\text{Number of Shutdowns}} \text{ (hours)}$$

$$\text{MTTR} = \frac{\text{Total Downtime Hours}}{\text{Number of Shutdowns}} \text{ (hours)}$$

$$\text{Impact on Availability} = (1 - \text{Availability}) \times \frac{\text{Downtime Hours}}{\text{Total Downtime Hours}} \times 100 \text{ (\%)}$$

$$\text{Cost per Hour (by system)} = \frac{\text{Cost}}{\text{Operating Hours}} \text{ (US$/hour)}$$

▪ PIP or PSP Completion

PIP or PSP Completion measures completeness of Product Improvement or Support Program execution during a period of time. The metric is used to monitor the status of implementation of factory programs. The factory program completion status is calculated as the ratio of programs completed on a machine-by-machine basis relative to the number of programs that are active and applicable at the time under consideration. This ratio is expressed as a percentage.

$$\text{PIP / PSP Completion} = \frac{\text{Factory Programs Completed}}{\text{Number of Programs that are Active}} \times 100 \text{ (\%)}$$

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Cost

Maintenance Cost is part of the total cost of operating a particular equipment and it is normally expressed in relation to the production figures of the equipment. As an example for hauling trucks it will be tons transported and for the loading tools tons loaded.

$$\text{Maintenance Cost} = \frac{\text{Total Maintenance Cost in the Period}}{\text{Tons}} \quad (\text{US\$})$$

M&R Outcomes and KPI's – Processes Outcome(s)

Now it is time to review the M&R processes outcome(s) or contributions to support the final equipment fleet performance.

Preventive Maintenance

“Contribute to equipment function preservation, availability and reliability by timely, efficient and effective execution of the PM services”

Specific Outcomes – Objectives

How do we **Measure** (KPI's)

PM services performed On Time	Service Accuracy
Efficient Execution	PM MTTR
Effective/Quality Execution	MTBS after PM
PM Availability Impact managed effectively	% of unavailability due to PM services
Minimum delays in the execution of the PM services	PM MTTR without Delays
Effective/efficient use of PM manpower	PM Maintenance Ratio
Effective Use of PM Window of Opportunity	Backlogs executed in PM Backlogs generated in PM

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Condition Monitoring (CM)

"Timely detection of abnormal conditions to support Failures Prevention"

Specific Outcomes – Objectives

How do we **Measure** (KPI's)

Effective Failures Prevention	MTBF Failure Reduction
Controlled Failures Impact	% unavailability due to Unscheduled Events
Effective CM Contribution / Effort	Total Backlogs Generated Backlogs generated by Origin
Effective Detection	Detection Level % of Failures without CM warning
Effective On Target Detection	% Backlogs generated in Top Ten Problems
Valuable CM Contribution	CM Cost Savings

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Backlog Management

“Efficient & Effective Management of Defects Detected”

Specific Outcomes – Objectives

How do we **Measure** (KPI's)

Backlog Pending effectively managed

Total Backlog Pending
Backlogs Pending per Machine

Backlog Balance in control

Backlogs Generated – Executed – Pending

Impact of Pending Backlogs in control

Estimated Time (DT) to Repair (ETTR) &
Estimated Labor to Repair (ELTR)

Backlog Management Process stages effectively managed

Backlog Status distribution

Avoid risk due to age of Backlogs

% of Backlogs > 30 days (aging)

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Planning & Scheduling

“Support the M&R operation with quality Plans & Instructions to contribute to meet the goal of 80% scheduled Down Time”

Specific Outcomes – Objectives

How do we **Measure** (KPI's)

Deliver Plans to gain the control of the equipment fleet:

- Plan for 80% scheduled Downtime
- Manage effectively the unavailability of equipment
- Maintain equipment Backlog in control

Timely schedule and execution of the M&R events / tasks

Plans and programs being followed and performed as scheduled

Components managed effectively

Timely & effective application of equipment factory updates/programs

Complete and accurate equipment repair history kept updated

% Scheduled Downtime
Scheduled Unavailability vs. Actual Unavailability
(Unavailability Distribution Analysis)
Backlog Management Results

Service Accuracy

Schedule Compliance by Hours
Scheduled Compliance by Events

Component Management Results
% of Components Exchanged Scheduled
% of Components > Life Target
ETTR of Components > Life Target

% of Factory Programs Completed
ETTR for Factory Programs not Completed

Work Order Management

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Repair Management

“Efficient (Time) & Effective (Quality) execution of Scheduled and Unscheduled M&R Events”

Specific Outcomes – Objectives

How do we **Measure** (KPI's)

Preparation and execution of the tasks managed effectively to meet the targeted down time	MTTR of Shop and Field MTTR without Delays Unavailability goals for various M&R categories Unavailability due to Delays
Quality execution of repairs and services	MTBS after short repairs & services Performance of equipment after repairs (MTBS & MTTR after 250 hours of big repairs completion) % of Redo
Effective support of the equipment reliability goals by using extensively the windows of opportunities	Backlogs executed in parallel Backlogs generated Backlog Execution Targets Completion
Effective use and guidance of Service Crew	Maintenance Ratio
Ensure Clean Work Environment	Contamination Control Ratings

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Parts Management

“Efficient & Effective Support to M&R activities: The Right Parts & Components and the Right Time and at Right Location”

Specific Outcomes - Objectives:

How do we **Measure** (KPI's)

Define and maintain updated the Parts & Component Inventory	Service Fill Level (Instantaneous) Service Fill Level (24 Hours)
Effective management of the Inventory	Inventory Items Inventory Cost Inventory Rotation
Efficient Response to Emergency Requests	Percentage of Emergency Orders Response Time for Emergency Orders
Support effectively the Backlog Management Program	% of Backlogs waiting for Parts Average Days for Backlogs waiting for Parts
Effective parts support to the overall M&R operation	% of Unavailability due to Parts

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Component Management

“Effective management of Components, a key Cost Driver”

Specific Outcomes – Objectives

How do we **Measure** (KPI's)

Achieve equipment component Life Target

% Component achieved Life Target

Component exchanged at the optimum time (Monitor the component closely to exchange them at the optimum time. Minimize Component Failures).

% Component Exchanged Scheduled

Monitor and manage the overall PCR to meet the PCR budget, the spinal cord of the CSA agreement.

Component Replacement Cost

Efficient execution of all component replacement maintaining the impact on the Availability in control.

Unavailability due to Component Replacement

PCR's executed with Quality to support the correct performance of the equipment after repair.
Effective use of the PCR window of Opportunity

Availability Index after Component Exchange

Parallel work (Backlogs) executed during Component Exchange

Planned Component Replacement Plan executed as planned

% PCR plan compliance

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HR / Training / HR Development

“Effective management of the most important Asset ... Our Personnel”

Specific Outcomes – Objectives

How do we **Measure** (KPI's)

Clear definition of the organization with clear R&R
Attract, hire, retain and manage the workforce efficiently.

Unavailability due to people / personnel Delays
Available / effective Man-Hours

Adequate quantity of personnel allocation to execute the M&R routines. Support them efficiently.

Hands On Personnel to Machine Ratio
Direct Maintenance Personnel
Total Personnel to Machine Ratio

Monitor and manage the total M&R operation personnel. Balance personnel needs and budget.

Total Maintenance Personnel

Manage the overtime within targets. Use overtime results as an indication of man-power availability and effectiveness. Always compare with % Unscheduled Work.

Overtime Ratio

Achieve optimum MR results.

Maintenance Ratio

Measure and provide adequate training to develop the people and to support the actual organization challenges

Training Hours Ratio
Training Hours by Systems
Training Program Compliance

Define and manage clearly the quantity of Competencies required and satisfied.

Competencies Required
Competencies Satisfied

Manage and Retain your people

Attrition Rate

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Continuous Improvement (CI)

“Effectively support the M&R operation with innovative, applicable and valuable solutions to allow the continuous search for excellence”

Specific Outcomes - Objectives

How do we **Measure** (KPI's)

Motivate the organization to constantly discovering new ideas to Improve.
Select & transform the ideas into CI projects
“Keep the Ideas flowing to the process”

CI Ideas Generated
CI Ideas to CI Projects Candidate

Keep the CI groups actively working (keep the blood flowing)

CI Projects Opened in the period
Active CI Projects

Guide the organization to the areas of “pain”

CI Projects to Top Ten Problems

Manage effectively the CI Initiatives

CI Projects On time
Average CI Project Execution
Projects Closed in the Period

Quantify the amount of valid solutions and value delivered to the organization. Manage accordingly.

CI Project Delivered Value
Quantity of Validated Projects Solution(s)

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Performance Evaluation

“Provide accurate and timely management information to support the effective control of the M&R processes”

In the case of the Performance Evaluation we need to support the following objectives. The achievement of these objectives will be measured by compliance.

- Provide a clear definition of the desired outcome(s) (What are we here for...) for the overall system and sub-processes
- Define together with respective process owners the Goals, Objectives and Targets for each particular process (Overall & Sub processes)
- Define the appropriate Key Performance Indicators to measure the defined Outcome(s)
- Identify the data needed to be collected to generate the KPI's and implement the Data Collection System
- Identify the Distribution Network and timely generate and communicate the KPI's and Management Reports
- Provide an accurate and effective interpretation, identifying the opportunities and correct actions to follow
- Identify the recommended corrective action(s) to follow and/or the need for Continuous Improvement in the immediate term and / or the need for the involvement of Continuous Improvement
- Follow up the results to validate or adjust the corrective or improvement action implemented. Keep Updated 3W plans.

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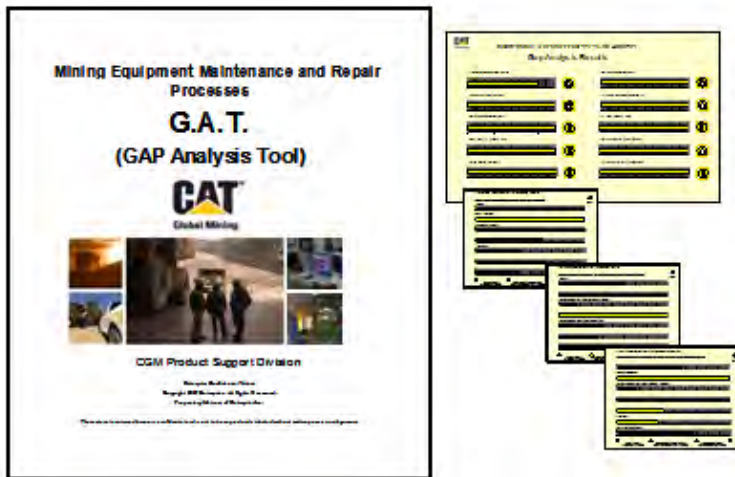
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3.0 Implementation Steps

The implementation steps to follow will depend on the current situation of your mine site operation. By saying that, what we are recommending as a first step is a thorough evaluation of your maintenance operation. Determine the “As Is” situation. Compare with the recommended model. Determine areas of opportunities and finally establish an action plan to correct or improve those areas of opportunities.

1. - Evaluate your on-site maintenance operation

- Apply GAT (Gap Analysis Tool) - available through Global Mining Product Support division



The GAT is a survey dedicated to evaluate the 10 different processes recommended by Caterpillar Global Mining.

2. - Establish the “As Is” situation

In establishing the “As Is” situation we recommend performing two distinct activities:

- Determine the Process Flow Diagrams of all your processes implemented on site

Based on the observations guided by the GAT tool and the on-site investigation draw and document the current processes on site.

- Establish the Baseline.

Measure the current performance results and document them for future reference. Apply the set of metrics recommended in CGM document dedicated to “KPI’s for evaluating Maintenance and Repair Processes”

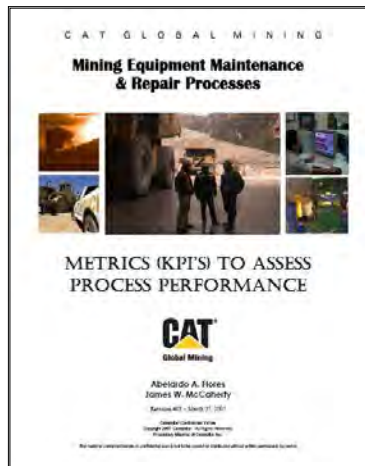
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This document contains the KPI's included in the GAT, calculation methodology and interpretation guidelines.

3. - Determine the areas of opportunities

Determine the areas of opportunities by comparing the “As Is” situation with the recommended processes detailed in the series of Strategic Best Practices (**BP 1007-2.0-1101 to BP 1007-2.0-1110**).

4. - Define the Action Plan

- Document your findings. The use of a 3 W form is recommended to record your findings, opportunities.
- Devise and document the solutions for improving or correcting the weaknesses. Use the related BP's to support your search for ideas / solutions. Use a separate column in your 3W form to record the solution / actions to take.
- Analyze and determine resources needed, time to implement and expected benefits.
- Prioritize the implementation of the different actions based on the Return on Investment (ROI) results.
- Assign responsibilities and expected completion dates for each of the activities.

5. - Get the “Buy-In” of the on-site organization

It is very important to get the buy-in of the on-site organization. They are the ultimate “process owners” of the maintenance & repair strategy. Present the action plan, the implementation strategy and the expected benefits. Involve them in the implementation of the solutions and plan an in-depth training of all the involved personnel.

6. - Implement

Execute the implementation plan.

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7. - Follow up

Follow up closely the implementation of your action plan.

- Set up “check points” / milestones
- Review compliance of committed dates for tasks completion
- Receive & analyze work progress from the specific responsible
- Reinforce the positive results
- Identify and eliminate threats and obstacles
- Support the ownership of the on-site organization
- Support the positive change of personnel, correct the negative attitudes or oppositions
- Make adjustments as necessary

8. - Measure the Results – Improve / Adjust if needed

It is absolutely necessary to measure the results of the improvements implemented at the level of each individual processes and at the final outcome, fleet performance.

Measure and trend the behavior of the KPI's selected for establishing the “baseline” when the “As Is” situation was established. Compare and adjust as necessary.

4.0 Benefits

“We can manage what we can measure”, this old and well known management principle summarize all benefits related to the Performance Evaluation M&R process.

We are confident that the implementation of the performance evaluation process together with the rest of the M&R processes and the disciplined execution of them will help you to gain the control of your maintenance processes. Gaining control of your maintenance operation is everything. If you are in control of the operation you can guide your organization and efforts to the direction your maintenance strategy and ultimately the production plan of the mine will indicate.

5.0 Resources Required

In general terms the resources needed are related to two key areas, people involved and in charge of the PE process and a data processing system to gather all captured data and transform it into valuable information.

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6.0 Supporting Attachments / References

The following documents are available to support the selection, use and interpretation of the recommended M&R KPI's.

Performance Metrics for Mobile Mining Equipment – rev 1.1.pdf
Mining Process Metrics - rev 03.pdf

7.0 Related Best Practices

CGM Cat M&R Process	BP 1007-2.0-1100
Preventive Maintenance	BP 1007-2.0-1101
Condition Monitoring	BP 1007-2.0-1102
Backlog Management	BP 1007-2.0-1103
Planning & Scheduling	BP 1007-2.0-1104
Parts Management	BP 1007-2.0-1105
Component Management	BP 1007-2.0-1106
Repair Management	BP 1007-2.0-1107
Human Resources / Training	BP 1007-2.0-1108
Performance Evaluation	BP 1007-2.0-1109
Continuous Improvement	BP 1007-2.0-1110

8.0 Acknowledgements

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